

Paper 1 - Set 01 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	D	Ratio, proportion and percentages
2	C	Number, units and estimation
3	F	Exponentials and logarithms
4	A	Exponentials and logarithms
5	E	Circles and circle theorems
6	A	Binomial expansion
7	B	Sequences and series
8	B	Indices and surds
9	E	Integration
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Worked solutions

Question 1 RAT-06 · Ratio, proportion and percentages · spec M3.10

Two similar solids have surface areas in the ratio 4 : 9. Find the ratio of their volumes.

Answer: D

Length ratio 2 : 3, so volume ratio $2^3 : 3^3 = 8 : 27$.

Full method - how and why

Similar solids: areas scale as the square of lengths, volumes as the cube. Work back to the length ratio first.

- 1 Areas are in ratio 4 : 9, so the length ratio is $\sqrt{4} : \sqrt{9} = 2 : 3$.

why: Area scales with the square of length, so undo it with a square root.

- 2 Volumes scale with the cube of length: $2^3 : 3^3$.

why: Every linear dimension is scaled, so three dimensions give the cube.

- 3 So the volume ratio is 8 : 27.

why: Check: the ratios 2 : 3, 4 : 9, 8 : 27 are the classic length/area/volume triple.

Where it goes wrong

- Cubing the **area** ratio to get 64 : 729 - an offered option.
- Leaving the answer as the length ratio 2 : 3.

Takeaway Length k , area k^2 , volume k^3 : always return to the length ratio before converting.

Question 2 NUM-03 · Number, units and estimation · spec M2.9

Express the recurring decimal $0.\dot{2}7$ as a fraction in its lowest terms.

Answer: C

$$x = 0.2727\dots, 100x = 27.2727\dots, \text{ so } 99x = 27 \text{ and } x = \frac{27}{99} = \frac{3}{11}.$$

Full method - how and why

The standard algebraic trick for recurring decimals: multiply by a power of 10 that shifts one full period, then subtract.

- 1 Let $x = 0.272727\dots$. The repeating block has **two** digits, so multiply by $10^2 = 100$: $100x = 27.272727\dots$

why: Shifting by exactly one period makes the decimal tails identical, so they cancel.

- 2 Subtract: $100x - x = 27.2727\dots - 0.2727\dots = 27$, so $99x = 27$.

why: The infinite tails cancel exactly - this is the whole method.

- 3 Solve: $x = \frac{27}{99}$.

why: The denominator is always $10^k - 1$, i.e. a string of nines matching the period length.

- 4 Simplify by 9: $\frac{3}{11}$.

why: The question demands lowest terms, and $\frac{27}{99}$ is offered as a trap.

Where it goes wrong

- Choosing $\frac{27}{99}$ - correct in value but not in lowest terms.
- Multiplying by 10 instead of 100, which does not align the tails.

Takeaway Multiply by $10^{(\text{period length})}$, subtract, then simplify fully.

Question 3 LOG-13 · Exponentials and logarithms · spec MM5.2

Given that $\log_{10} 2 = p$ and $\log_{10} 3 = q$, express $\log_{10} 0.75$ in terms of p and q .

Answer: F

$$0.75 = \frac{3}{4}, \text{ so } \log 0.75 = \log 3 - \log 4 = q - 2p.$$

Full method - how and why

Express the number in terms of the given primes, then apply the log laws.

1 Write $0.75 = \frac{3}{4}$.

why: Decimals must become fractions of small numbers before the laws help.

2 Split: $\log_{10} \frac{3}{4} = \log_{10} 3 - \log_{10} 4$.

why: Quotient law.

3 Write $4 = 2^2$: $\log_{10} 4 = 2\log_{10} 2 = 2p$.

why: Only $\log 2$ and $\log 3$ are available, so every number must be built from 2s and 3s.

4 Combine: $q - 2p$.

why: Substituting $\log_{10} 3 = q$.

Where it goes wrong

- Using $\log 4 = p^2$ or $= 2 + p$.
- Getting the subtraction backwards and answering $2p - q$.

Takeaway Factorise the argument into the primes you have been given, then apply product/quotient/power laws.

Question 4 LOG-10 · Exponentials and logarithms · spec MM5.3

How many real solutions does the equation $4^x + 2^{x+1} - 8 = 0$ have?

Answer: A

$$y = 2^x: y^2 + 2y - 8 = (y + 4)(y - 2) = 0. y = -4 \text{ is impossible, } y = 2 \text{ gives } x = 1. \text{ One solution.}$$

Full method - how and why

Quadratic in 2^x again, but this time one root must be rejected - which changes the count.

1 Rewrite: $4^x = (2^2)^x = (2^x)^2$ and $2^{x+1} = 2 \cdot 2^x$.

why: Get everything in terms of 2^x before substituting.

2 Let $y = 2^x$: $y^2 + 2y - 8 = 0$, which factorises as $(y + 4)(y - 2) = 0$.

why: Standard quadratic.

3 Reject $y = -4$: 2^x is positive for every real x , so it can never equal a negative number.

why: This is the crucial filter - exponentials are always strictly positive.

4 From $y = 2$: $2^x = 2$, so $x = 1$. Exactly **one** real solution.

why: The question asks how many, so the count is the answer.

Where it goes wrong

- Answering 2 because the quadratic has two roots.
- Rejecting $y = 2$ as well, and answering 0.

Takeaway After substituting $y = a^x$, discard every root with $y \leq 0$ - then count what remains.

Question 5 CIR-14 · Circles and circle theorems · spec MM3.3

AB is a diameter of a circle and C is a point on the circle with $AC = 6$ and $BC = 8$. Find the radius of the circle.

Answer: E

The angle in a semicircle is a right angle, so $AB = \sqrt{36 + 64} = 10$ and the radius is 5.

Full method - how and why

The angle in a semicircle is a right angle, so AB is the hypotenuse of a right-angled triangle.

- 1 AB is a diameter and C is on the circle, so angle $ACB = 90^\circ$.
why: The angle in a semicircle is a right angle - the key theorem here.
- 2 Apply Pythagoras: $AB = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$.
why: The right angle is at C , so the diameter AB is the hypotenuse.
- 3 Radius $= \frac{10}{2} = 5$.
why: The question asks for the radius, not the diameter.

Where it goes wrong

- Answering 10 (the diameter).
- Assuming the right angle is at A or B .

Takeaway A triangle inscribed with one side a diameter is right-angled at the third vertex.

Question 6 BIN-02 · Binomial expansion · spec MM2.4

The coefficient of x^2 in the expansion of $(1 + kx)^6$ is 60. Given that $k > 0$, find k .

Answer: A

$$\binom{6}{2}k^2 = 15k^2 = 60, \text{ so } k^2 = 4 \text{ and } k = 2.$$

Full method - how and why

A coefficient is given and a constant inside the bracket is unknown. Set up the coefficient equation and solve.

- 1 General term of $(1 + kx)^6$: $\binom{6}{r}(kx)^r$, so the x^2 coefficient is $\binom{6}{2}k^2$.
why: The 1^{6-r} factor is invisible, which is why this form is the easiest to work with.
- 2 Compute $\binom{6}{2} = 15$, so the equation is $15k^2 = 60$.
why: $\binom{6}{2} = \frac{6 \times 5}{2} = 15$.
- 3 Solve: $k^2 = 4$, so $k = \pm 2$; the condition $k > 0$ gives $k = 2$.
why: Squaring means two roots; the stated condition selects one.

Where it goes wrong

- Forgetting to square k and solving $15k = 60$ to get $k = 4$ - which is offered.
- Using $\binom{6}{2} = 30$ (that is 6×5 without dividing by 2).

Takeaway The x^r coefficient of $(1 + kx)^n$ is $\binom{n}{r}k^r$ - the constant is raised to the same power as x .

Question 7 SEQ-14 · Sequences and series · spec MM2.1

The sequence $u_n = n^2 - 10n + 30$ is defined for positive integers n . What is the smallest value taken by any term of the sequence?

Answer: B

$$u_n = (n - 5)^2 + 5, \text{ minimised at } n = 5 \text{ with value } 5.$$

Full method - how and why

A quadratic sequence. Completing the square finds the minimum - but n must be a positive integer, so check that the optimum is attainable.

- 1 Complete the square: $n^2 - 10n + 30 = (n - 5)^2 + 5$.

why: Half of -10 is -5 ; $30 - 25 = 5$.

- 2 The square is smallest when $n = 5$, giving $u_5 = 5$.

why: $(n - 5)^2 \geq 0$ with equality only at $n = 5$.

- 3 $n = 5$ is a positive integer, so the minimum is attained: the smallest term is 5.

why: If the vertex had been at a non-integer (say 4.5), you would have to test the integers either side.

Where it goes wrong

- Giving $n = 5$ (the position) instead of 5 (the value) - here they coincide, which hides the error.
- Taking $u_1 = 21$ as the smallest by assuming sequences always increase.

Takeaway Minimise a quadratic sequence by completing the square, then check the nearest **integer** values of n .

Question 8 IND-08 · Indices and surds · spec MM1.1

Solve $9^x - 4 \times 3^x + 3 = 0$.

Answer: B

$$\text{Let } y = 3^x: y^2 - 4y + 3 = (y - 1)(y - 3) = 0, \text{ so } 3^x = 1 \text{ or } 3^x = 3, \text{ giving } x = 0 \text{ or } x = 1.$$

Full method - how and why

A hidden quadratic. The key observation is $9^x = (3^2)^x = (3^x)^2$, so the equation is quadratic in 3^x .

- 1 Write $9^x = (3^x)^2$ and substitute $y = 3^x$: $y^2 - 4y + 3 = 0$.

why: Spotting that one exponential is the square of the other is the whole question; without it there is no route in.

- 2 Factorise: $(y - 1)(y - 3) = 0$, so $y = 1$ or $y = 3$.

why: Ordinary quadratic technique once the substitution is made.

- 3 Undo the substitution: $3^x = 1 \Rightarrow x = 0$; $3^x = 3 \Rightarrow x = 1$.

why: $a^0 = 1$ for any a - the root $y = 1$ is easy to lose because it looks trivial.

Where it goes wrong

- Stopping at $y = 1, 3$ and offering those as the answers - the question asks for x , not 3^x .
- Rejecting $y = 1$ as 'not a real solution'. Only **non-positive** values of y must be rejected, because $3^x > 0$ always.

Takeaway If an equation contains a^{2x} and a^x , substitute $y = a^x$; afterwards discard only the roots with $y \leq 0$.

Question 9 INT-13 · Integration · spec MM7.3

$f(x) = ax + b$ satisfies $\int_0^1 f(x) dx = 2$ and $\int_0^2 f(x) dx = 6$. Find $f(1)$.

Answer: E

$\frac{a}{2} + b = 2$ and $2a + 2b = 6$, so $a + b = 3$; subtracting the first equation from this gives $\frac{a}{2} = 1$, so $a = 2$, $b = 1$ and $f(1) = 3$.

Full method - how and why

Two integral conditions on a linear function give two equations in a and b .

- 1 Integrate in general: $\int_0^k (ax + b) dx = \frac{ak^2}{2} + bk$.

why: Doing it once with a general limit saves repeating the work.

- 2 $k = 1$: $\frac{a}{2} + b = 2$. $k = 2$: $2a + 2b = 6$, i.e. $a + b = 3$.

why: Substitute each limit and simplify the second by dividing by 2.

- 3 Subtract the first from the second: $\frac{a}{2} = 1$, so $a = 2$, and then $b = 1$.

why: Elimination: $a - \frac{a}{2} = \frac{a}{2}$ and $3 - 2 = 1$.

- 4 Compute $f(1) = a + b = 3$.

why: Conveniently, $f(1) = a + b$, which the second equation already gave.

Where it goes wrong

- Forgetting the $\frac{1}{2}$ in $\frac{ak^2}{2}$ and getting a different pair.
- Answering 2 (the value of the first integral) instead of $f(1)$.

Takeaway Integrate once with a general limit, then substitute each condition to get simultaneous equations.

Question 10 SIM-05 · Simultaneous equations and inequalities · spec MM1.5

For how many integers n is $n^2 - 10n + 16 < 0$?

Answer: A

$(n - 2)(n - 8) < 0 \Rightarrow 2 < n < 8$, so $n = 3, 4, 5, 6, 7$: five integers.

Full method - how and why

Counting integers in the solution set of a quadratic inequality. The counting at the end is where the marks are won or lost.

- 1 Factorise: $n^2 - 10n + 16 = (n - 2)(n - 8)$.

why: Two numbers multiplying to 16, adding to -10 : -2 and -8 .

- 2 Negative between the roots: $2 < n < 8$.

why: Upward parabola, so it dips below the axis strictly between its roots.

- 3 List the integers strictly inside: 3, 4, 5, 6, 7 - five values.

why: Strict inequality excludes 2 and 8, where the expression equals zero.

Where it goes wrong

- Including the endpoints and answering 7.
- Counting $8 - 2 = 6$ as the number of integers - the count between a and b exclusive is $b - a - 1$.

Takeaway Write the integers out; the number strictly between a and b is $b - a - 1$, not $b - a$.

Question 11 TRG-18 · Trigonometric functions and equations · spec MM4.3

Find the value of $\tan 30^\circ \tan 60^\circ + \sin^2 45^\circ$.

Answer: C

$$\tan 30^\circ \tan 60^\circ = \frac{1}{\sqrt{3}} \cdot \sqrt{3} = 1, \sin^2 45^\circ = \frac{1}{2}. \text{ Total } \frac{3}{2}.$$

Full method - how and why

Exact-value recall plus one squared value. The first product is designed to collapse to 1.

1 $\tan 30^\circ = \frac{1}{\sqrt{3}}$ and $\tan 60^\circ = \sqrt{3}$, so their product is 1.
why: 30° and 60° are complementary, and $\tan \theta \tan(90^\circ - \theta) = 1$ always.

2 $\sin 45^\circ = \frac{\sqrt{2}}{2}$, so $\sin^2 45^\circ = \frac{2}{4} = \frac{1}{2}$.
why: Square top and bottom: $(\sqrt{2})^2 = 2$ and $2^2 = 4$.

3 Add: $1 + \frac{1}{2} = \frac{3}{2}$.
why: Straightforward once both pieces are exact.

Where it goes wrong

- Writing $\sin^2 45^\circ = \frac{\sqrt{2}}{2}$ (forgetting to square), giving $1 + \frac{\sqrt{2}}{2}$.
- Using $\tan 30^\circ = \sqrt{3}$, which makes the product 3.

Takeaway $\tan \theta \tan(90^\circ - \theta) = 1$, and $\sin^2 45^\circ = \cos^2 45^\circ = \frac{1}{2}$.

Question 12 ALG-06 · GCSE algebra · spec M4.3

Given that $v^2 = u^2 + 2as$, find s when $v = 13$, $u = 5$ and $a = 4$.

Answer: A

$$169 = 25 + 8s \Rightarrow s = 18.$$

Full method - how and why

Substitution into a formula. Evaluate the squares first, then solve the linear equation.

1 Substitute: $13^2 = 5^2 + 2(4)s$, i.e. $169 = 25 + 8s$.
why: Compute each square separately; $2as = 2 \times 4 \times s = 8s$.

2 Rearrange: $8s = 144$.
why: $169 - 25 = 144$.

3 Divide: $s = 18$.
why: $144 \div 8 = 18$.

Where it goes wrong

- Forgetting the factor 2 in $2as$ and getting 36 - an offered option.
- Computing $13^2 = 139$ or $5^2 = 10$.

Takeaway Substitute numbers for every letter, evaluate powers first, then solve.

Question 13 TRI-01 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , $AB = 5$, $AC = 8$ and angle $BAC = 60^\circ$. Find BC .

Answer: F

Cosine rule: $BC^2 = 25 + 64 - 2 \cdot 5 \cdot 8 \cdot \frac{1}{2} = 89 - 40 = 49$, so $BC = 7$.

Full method - how and why

Two sides and the angle between them: that is exactly the cosine rule's input.

- 1 Choose the rule: you know two sides and the **included** angle, so use the cosine rule, not the sine rule.
why: The sine rule needs a side opposite a known angle; here the known angle sits between the known sides.
- 2 Write it with BC opposite the known angle A : $BC^2 = AB^2 + AC^2 - 2(AB)(AC)\cos A$.
why: The side being found must be opposite the angle being used.
- 3 Substitute, using $\cos 60^\circ = \frac{1}{2}$: $BC^2 = 25 + 64 - 2(5)(8)(\frac{1}{2}) = 89 - 40 = 49$.
why: $\cos 60^\circ = \frac{1}{2}$ is one of the exact values you must recall.
- 4 Take the square root: $BC = 7$.
why: Lengths are positive, so only the positive root.

Where it goes wrong

- Forgetting the $-2bc\cos A$ term and answering $\sqrt{89}$ - which is offered.
- Using $\cos 60^\circ = \frac{\sqrt{3}}{2}$ (that is $\cos 30^\circ$), giving $\sqrt{89 - 40\sqrt{3}}$.

Takeaway Two sides plus the included angle $\Rightarrow$ cosine rule; the unknown side is opposite the known angle.

Question 14 SIM-02 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of x for which $2x^2 - 5x - 3 \geq 0$.

Answer: A

$(2x + 1)(x - 3) \geq 0$: the roots are $-\frac{1}{2}$ and 3 and the parabola opens upwards, so $x \leq -\frac{1}{2}$ or $x \geq 3$.

Full method - how and why

A quadratic inequality with a leading coefficient other than 1. Factorise, find the roots, then use the shape of the parabola.

- 1 Factorise: $2x^2 - 5x - 3 = (2x + 1)(x - 3)$.
why: Two numbers multiplying to $2 \times (-3) = -6$ and adding to -5 : -6 and 1 , so split $-5x$ as $-6x + x$.
- 2 Roots: $x = -\frac{1}{2}$ and $x = 3$.
why: Set each bracket to zero.
- 3 The coefficient of x^2 is positive, so the parabola opens upwards and is ≥ 0 **outside** the roots.
why: A quick sketch is the safest way to convert a factorisation into an inequality answer.
- 4 Answer: $x \leq -\frac{1}{2}$ or $x \geq 3$, with $\leq$ because equality is allowed.
why: The question has ≥ 0 , so the roots themselves are included.

Where it goes wrong

- Giving $-\frac{1}{2} \leq x \leq 3$, which is where the expression is negative.
- Mixing up the roots as -3 and $\frac{1}{2}$ - the root of $(2x + 1)$ is $-\frac{1}{2}$, not -2 .

Takeaway Factorise, mark the roots, sketch the parabola: 'outside' for > 0 , 'between' for < 0 when $a > 0$.

Question 15 SEQ-05 · Sequences and series · spec MM2.3

A geometric series has first term a and common ratio r with $|r| < 1$. Its sum to infinity is 8 and the sum to infinity of the squares of its terms is $\frac{64}{3}$. Find r .

Answer: B

$$\frac{a}{1-r} = 8 \text{ and } \frac{a^2}{1-r^2} = \frac{64}{3}. \text{ Dividing the second by the square of the first: } \frac{1-r}{1+r} = \frac{64/3}{64} = \frac{1}{3}, \text{ so } 3 - 3r = 1 + r \text{ and } r = \frac{1}{2}.$$

Full method - how and why

Two infinite sums, two unknowns. Dividing one equation by the (square of the) other eliminates a elegantly.

- Write the two facts: $\frac{a}{1-r} = 8$ and, for the squares (first term a^2 , ratio r^2), $\frac{a^2}{1-r^2} = \frac{64}{3}$.
why: Squaring every term of a geometric series gives another geometric series with first term a^2 and ratio r^2 .
- Square the first: $\frac{a^2}{(1-r)^2} = 64$.
why: Squaring makes the numerators match, so they can be eliminated by division.
- Divide the second equation by this: $\frac{(1-r)^2}{1-r^2} = \frac{64/3}{64} = \frac{1}{3}$.
why: Dividing kills a^2 entirely - the standard move when two equations share a factor.
- Simplify using $1-r^2 = (1-r)(1+r)$: $\frac{1-r}{1+r} = \frac{1}{3}$.
why: One factor of $(1-r)$ cancels.
- Cross-multiply: $3 - 3r = 1 + r \Rightarrow 4r = 2 \Rightarrow r = \frac{1}{2}$.
why: Check: $a = 4$, and $\frac{16}{1-1/4} = \frac{64}{3}$ ✓.

Where it goes wrong

- Using ratio r (not r^2) for the series of squares.
- Forgetting to square the first equation before dividing, which leaves an a behind.

Takeaway Squaring a geometric series squares both a and r ; divide equations to eliminate the unknown you do not want.

Question 16 BIN-03 · Binomial expansion · spec MM2.4

Find the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.

Answer: E

General term $\binom{6}{r}(x^2)^{6-r}\left(\frac{2}{x}\right)^r = \binom{6}{r}2^r x^{12-3r}$. Need $12 - 3r = 0$, $r = 4$: $\binom{6}{4} \times 16 = 15 \times 16 = 240$.

Full method - how and why

'Term independent of x ' means the power of x is zero. Track the power of x in the general term and solve for r .

- General term: $\binom{6}{r}(x^2)^{6-r}\left(\frac{2}{x}\right)^r = \binom{6}{r}2^r x^{12-2r-r}$.
why: $(x^2)^{6-r}$ gives x^{12-2r} and $\left(\frac{2}{x}\right)^r$ gives $2^r x^{-r}$; combine the powers.
- So the power of x is $12 - 3r$.
why: Getting this expression right is the whole question.
- Set it to zero: $12 - 3r = 0$, so $r = 4$.
why: 'Independent of x ' means x^0 .
- Evaluate the term: $\binom{6}{4} \times 2^4 = 15 \times 16 = 240$.
why: $\binom{6}{4} = \binom{6}{2} = 15$ by symmetry.

Where it goes wrong

- Using the power $12 - 2r$ and getting $r = 6$, giving $2^6 = 64$.
- Forgetting the 2^r factor and answering 15.

Takeaway Write the power of x in the general term as a linear expression in r , then set it to the value you need.

Question 17 LIN-03 · Straight lines · spec MM3.1

The line L passes through $(4, 0)$ and $(0, 6)$. Find the perpendicular distance from the origin to L .

Answer: D

Area of triangle $= \frac{1}{2} \cdot 4 \cdot 6 = 12$. Hypotenuse $= \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$. Distance d satisfies $\frac{1}{2} \cdot 2\sqrt{13} \cdot d = 12$, so $d = \frac{12}{\sqrt{13}}$.

Full method - how and why

Perpendicular distance from a point to a line. The 'area two ways' method avoids memorising the distance formula.

- 1 The line cuts the axes at $(4, 0)$ and $(0, 6)$, so with the origin it forms a right-angled triangle of area $\frac{1}{2} \times 4 \times 6 = 12$.

why: The two intercepts and the origin make a triangle whose legs lie along the axes - its area is easy.

- 2 The hypotenuse (the segment of L) has length $\sqrt{4^2 + 6^2} = \sqrt{52} = 2\sqrt{13}$.

why: Pythagoras on the same triangle.

- 3 Compute the area again using the hypotenuse as base and the required distance d as height: $\frac{1}{2} \times 2\sqrt{13} \times d = 12$.

why: The perpendicular distance from the origin to L is the height of that triangle on that base.

- 4 Solve: $d = \frac{12}{\sqrt{13}}$.

why: Equivalently, the formula $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$ with $3x + 2y = 12$ gives $\frac{12}{\sqrt{13}}$.

Where it goes wrong

- Using the distance to one of the intercepts $(4, 6 \text{ or } 2\sqrt{13})$ instead of the perpendicular distance.
- Forgetting the square root and answering $\frac{12}{13}$.

Takeaway Area computed two ways turns a distance problem into arithmetic; or use $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

Question 18 GRA-07 · Graphs of functions · spec MM8.5

The graph of $y = f(x)$ has $f'(x) > 0$ for $x < 1$, $f'(1) = 0$, $f'(x) > 0$ for $1 < x < 3$, $f'(x) < 0$ for $x > 3$. Which of the following is true?

Answer: C

At $x = 1$ the derivative is zero but does not change sign: a stationary point that is not a turning point (an inflexion).

At $x = 3$ the derivative changes from positive to negative: local maximum.

Full method - how and why

Reading a function's shape from the sign of its derivative. A zero derivative **without** a sign change is an inflexion, not a turning point.

- 1 Before and after $x = 1$: $f' > 0$ on both sides, with $f'(1) = 0$.

why: The gradient dips to zero and comes back positive - the curve flattens momentarily but keeps rising.

- 2 So $x = 1$ is a stationary point that is **not** a turning point: a stationary point of inflexion.

why: Turning points require a change of sign in f' .

- 3 At $x = 3$: $f' > 0$ just before and $f' < 0$ just after, so the curve rises then falls - a local maximum.

why: A $+$ $\rightarrow$ $-$ sign change is exactly a local maximum.

- 4 So: local maximum at $x = 3$, stationary inflexion at $x = 1$.

why: Match to the options.

Where it goes wrong

- Assuming $f'(1) = 0$ must be a maximum or minimum.
- Reading the $+$ $\rightarrow$ $-$ change at $x = 3$ as a minimum.

Takeaway $f' = 0$ with no sign change = inflexion; $+$ $\rightarrow$ $-$ is a maximum, $- \rightarrow +$ is a minimum.

Question 19 IND-09 · Indices and surds · spec MM1.2

Find the value of $\frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{1}{2+\sqrt{3}}$.

Answer: D

Rationalising each term gives $(\sqrt{2}-1) + (\sqrt{3}-\sqrt{2}) + (2-\sqrt{3})$, which telescopes to 1.

Full method - how and why

A telescoping sum. Each term rationalises to a difference of two surds, and almost everything cancels - so do not compute decimals.

1 Rationalise each term: $\frac{1}{\sqrt{2}+1} = \sqrt{2}-1$, $\frac{1}{\sqrt{3}+\sqrt{2}} = \sqrt{3}-\sqrt{2}$, $\frac{1}{2+\sqrt{3}} = 2-\sqrt{3}$.

why: Each denominator has the form $\sqrt{n+1} + \sqrt{n}$ whose conjugate gives $(\sqrt{n+1} + \sqrt{n})(\sqrt{n+1} - \sqrt{n}) = 1$ - the denominators all become 1, which is the point of the construction.

2 Add them: $(\sqrt{2}-1) + (\sqrt{3}-\sqrt{2}) + (2-\sqrt{3})$.

why: Write the terms in a line so the cancelling pairs are visible.

3 Cancel: $\sqrt{2}$ with $-\sqrt{2}$, $\sqrt{3}$ with $-\sqrt{3}$, leaving $-1 + 2 = 1$.

why: This is a telescoping sum - the surds annihilate in pairs and only the two ends survive.

Where it goes wrong

- Approximating numerically and choosing $\sqrt{2}$ or $\sqrt{3}$ because the total 'looks about right'.
- Rationalising with the wrong conjugate sign, which flips one term and destroys the cancellation.

Takeaway $\frac{1}{\sqrt{n+1} + \sqrt{n}} = \sqrt{n+1} - \sqrt{n}$: recognise this and long chains of such fractions collapse instantly.

Question 20 PRO-08 · Probability · spec M7.7

A box contains n white balls and 3 black balls. Two balls are drawn without replacement. The probability that both are white is $\frac{5}{12}$. Find n .

Answer: A

$$\frac{n}{n+3} \cdot \frac{n-1}{n+2} = \frac{5}{12} \Rightarrow 12n^2 - 12n = 5n^2 + 25n + 30 \Rightarrow 7n^2 - 37n - 30 = 0 \Rightarrow (7n+5)(n-6) = 0, \text{ so } n = 6.$$

Full method - how and why

Set up the without-replacement probability in terms of n , then solve the resulting quadratic.

1 There are $n+3$ balls in total. $P(\text{both white}) = \frac{n}{n+3} \times \frac{n-1}{n+2}$.

why: After one white is removed, $n-1$ whites remain out of $n+2$.

2 Set equal to $\frac{5}{12}$ and cross-multiply: $12n(n-1) = 5(n+3)(n+2)$.

why: Cross-multiplying is safe since all denominators are positive.

3 Expand: $12n^2 - 12n = 5(n^2 + 5n + 6) = 5n^2 + 25n + 30$.

why: Expand the right side carefully: $(n+3)(n+2) = n^2 + 5n + 6$.

4 Collect: $7n^2 - 37n - 30 = 0$, which factorises as $(7n+5)(n-6) = 0$.

why: Check the middle term: $-42n + 5n = -37n$ ✓.

5 $n = -\frac{5}{7}$ is impossible for a count, so $n = 6$.

why: Verify: $\frac{6}{9} \times \frac{5}{8} = \frac{30}{72} = \frac{5}{12}$ ✓.

Where it goes wrong

- Writing $(\frac{n}{n+3})^2$, the with-replacement version.
- Accepting the negative root instead of rejecting it on contextual grounds.

Takeaway Model without replacement by decreasing both numerator and denominator, then reject roots that cannot be counts.